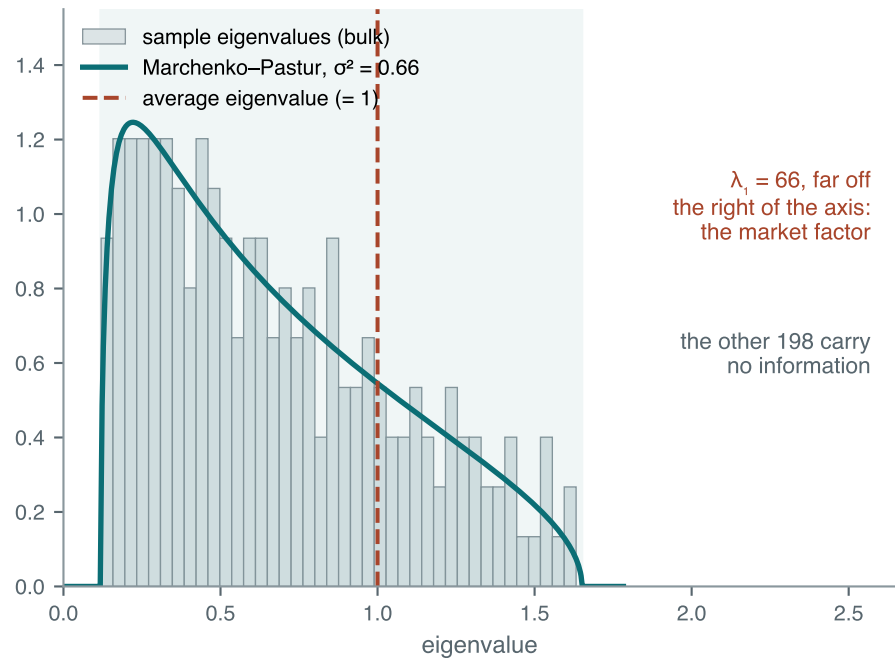
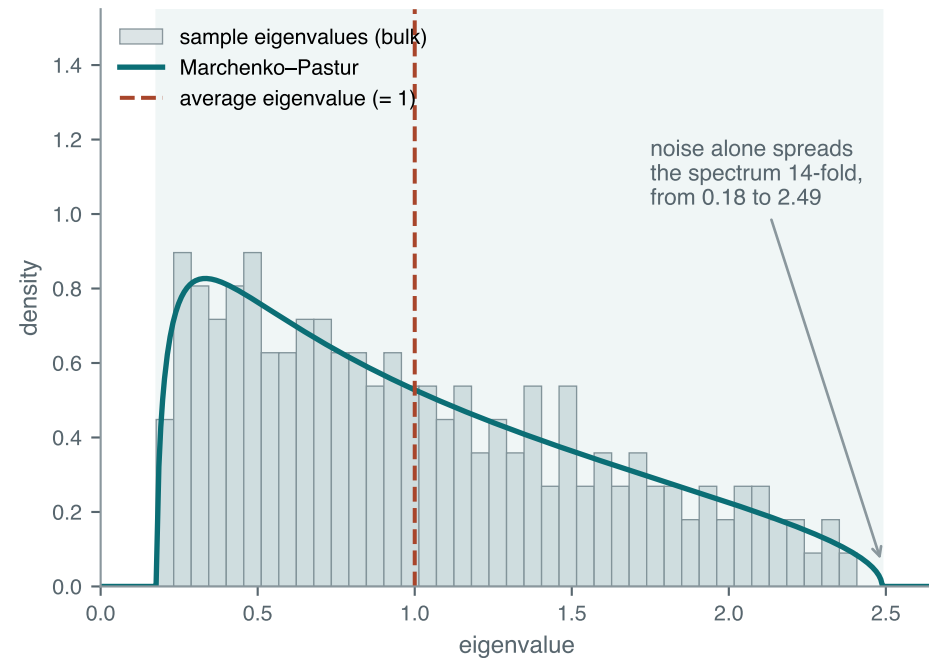


No structure — every eigenvalue is noise

One market factor — a spike, and a noise bulk



$N = 200$ assets, $T = 600$ observations, $q = N/T = 0.33$. Left: the truth is a single eigenvalue at 1, and the sample smears it across the shaded band. Right: once the market mode's share of the trace is taken out, what remains still matches the same law — which is exactly what makes it safe to discard.